How does data structure shape the network?

$$\text{tensor} \rightarrow \text{locality} \rightarrow \text{convolution} \rightarrow \text{hierarchy} \rightarrow \text{transfer} \rightarrow \text{patches}$$

Roadmap

1. Images are structured data — what an MLP ignores
2. Convolution — locality and weight sharing as a layer
3. CNNs and residual networks — stacking local layers into a hierarchy
4. Transfer learning — reusing features learned on other data
5. From patches to attention — a weaker assumption, more data

Each part names one **assumption** about images and shows the evidence for when it helps.

From deep training to architecture

Session 02 asked whether a deep f_θ **can be trained**.

Session 03 asks **which** f_θ to train.

$$x \to \underbrace{f_\theta(x)}_{\text{now with structure}} \to \hat{y} \to L \to \nabla_\theta L \to \theta'$$

The loop, the optimizers and the diagnostics stay the same.

The question for today

> **What assumption does this architecture introduce, and what evidence shows whether that assumption helps in this problem?**

An architecture is a bet about the data.

A good bet saves parameters and data. A wrong bet costs accuracy that no optimizer can recover.

Part 1 · Images are structured data

1. **Images are structured data** — what an MLP ignores
2. Convolution — locality and weight sharing as a layer
3. CNNs and residual networks — stacking local layers into a hierarchy
4. Transfer learning — reusing features learned on other data
5. From patches to attention — a weaker assumption, more data

> Session 02: we can train deep networks. Now: what should they look like?

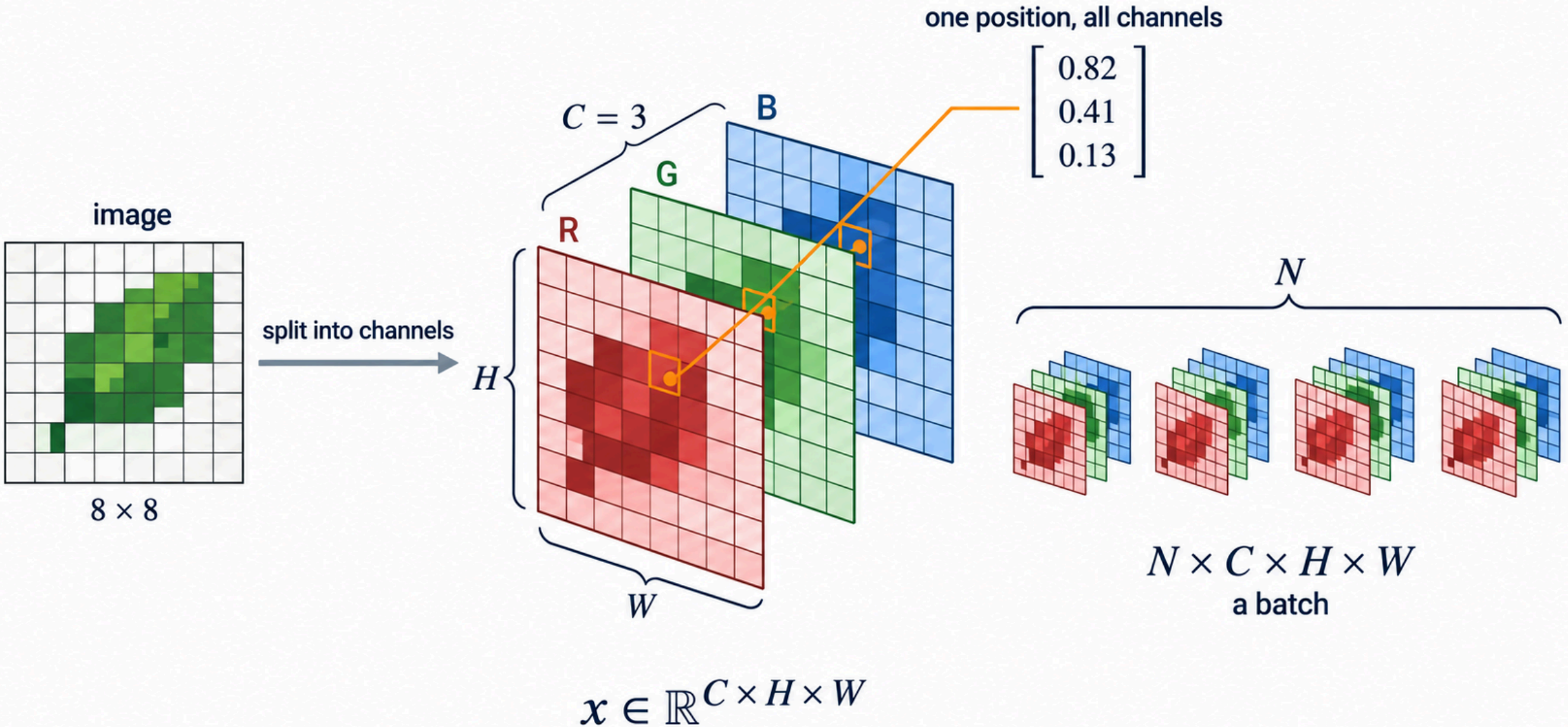
An image is a tensor

A color image is a 3-D array:

$$x \in \mathbb{R}^{C \times H \times W}, \qquad C = 3 \ (\mathrm{RGB})$$

A batch adds one dimension: $N \times C \times H \times W$.

Values are usually scaled to $[0, 1]$ and then normalized **per channel**.



$$x \in \mathbb{R}^{C \times H \times W}$$

Flattening an image into an MLP

CIFAR-10: $3 \times 32 \times 32 = 3072$ inputs.

A first hidden layer with 512 units has

$$3072 \times 512 + 512 \approx 1.6\ \mathrm{M}\ \text{parameters}$$

For a 224×224 image the same layer needs about **77 M** parameters.

What flattening discards

After `flatten`, the MLP sees a vector of 3072 numbers.

Nothing tells it that pixel 1 and pixel 2 are **neighbors**, or that pixel 1 and pixel 33 are **one row apart**.

Every input position gets its own weights: a pattern learned in one corner must be learned again in the others.

Prediction before experiment

Choose one fixed random permutation π of the 3072 positions.

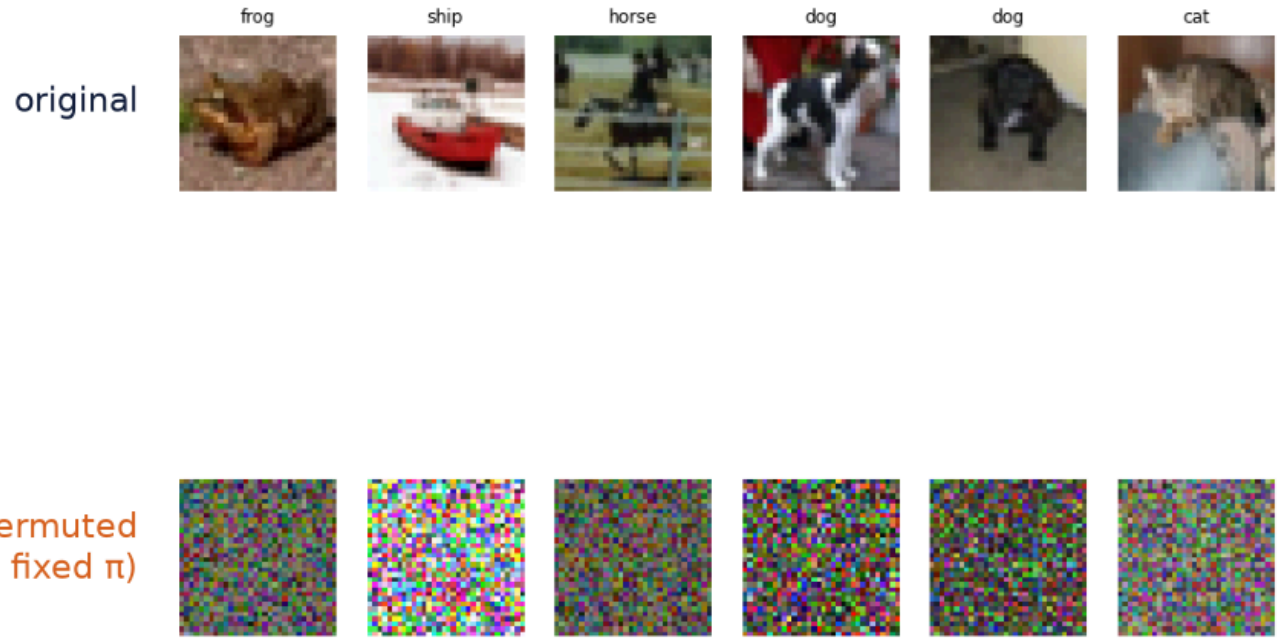
Apply it to **every** image, in training and in test.

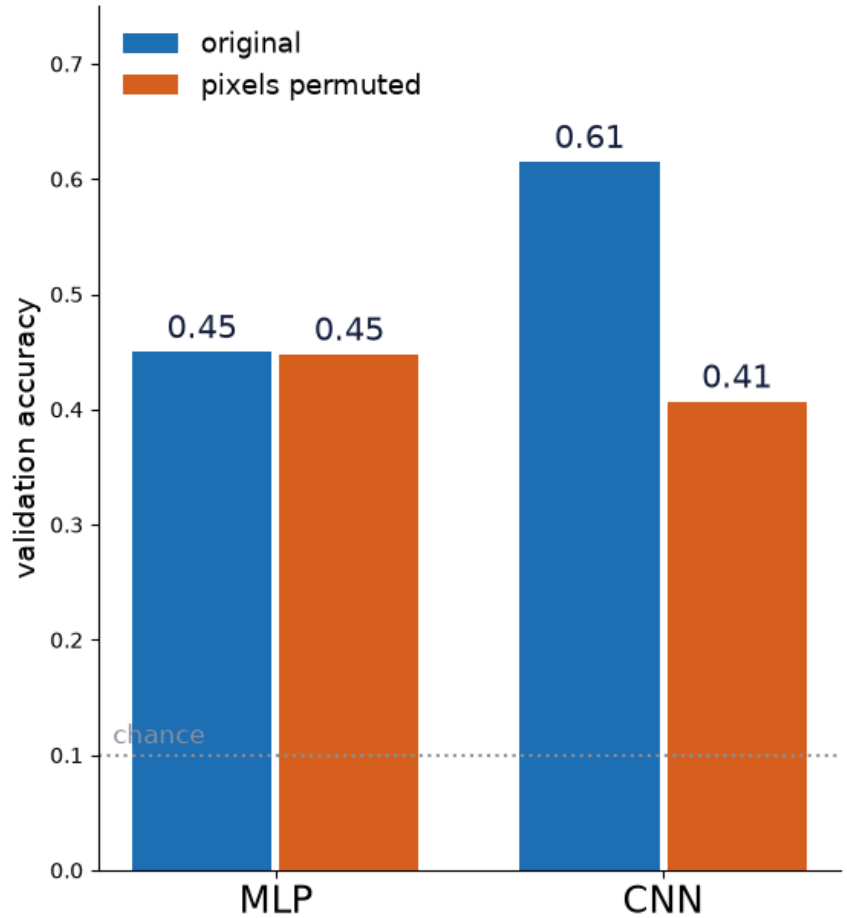
MLP trained on permuted images:

accuracy goes up, down, or stays the same?

CNN trained on permuted images:

accuracy goes up, down, or stays the same?





The MLP does not notice the permutation. The CNN loses its advantage: it assumed that neighbors matter.

Two assumptions about images

Locality. Nearby pixels are strongly related; distant pixels much less.

Translation equivariance. A pattern means the same thing wherever it appears: an edge is an edge in any corner.

An MLP assumes neither. A convolutional layer assumes both.

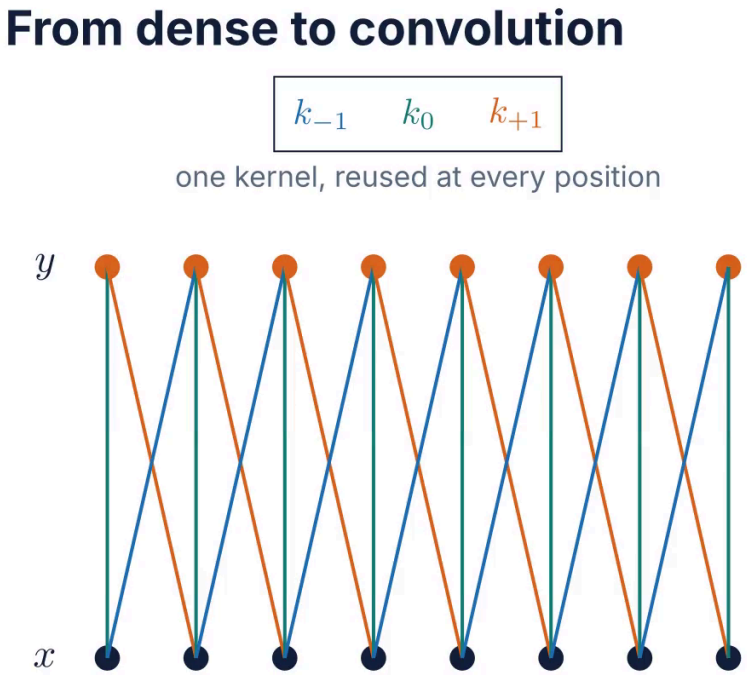
Dense, local, shared

For a 32×32 single-channel input mapped to a 32×32 output:

Layer	Each output sees	Parameters
Dense	all 1024 inputs	$\approx 1\,050\,000$
Locally connected, 3×3	9 neighbors, own weights	$\approx 10\,200$
Convolution, 3×3	9 neighbors, **shared** weights	10

Locality removes most connections; sharing removes most of the rest.

From dense to convolution



parameters (with bias)

	drawn	32 × 32 image
dense	8·8 + 8 = 72	**1 049 600**
local, 3 wide	8·3 + 8 = 32	**10 240**
shared kernel	3 + 1 = 4	**10**

Sharing: the same 3 weights at every position. That is a convolution.

Open animation

Part 2 · Convolution

1. Images are structured data — what an MLP ignores
2. **Convolution** — locality and weight sharing as a layer
3. CNNs and residual networks — stacking local layers into a hierarchy
4. Transfer learning — reusing features learned on other data
5. From patches to attention — a weaker assumption, more data

> Part 1: the MLP ignores neighborhoods; locality and sharing reduce 1 M parameters to 10.

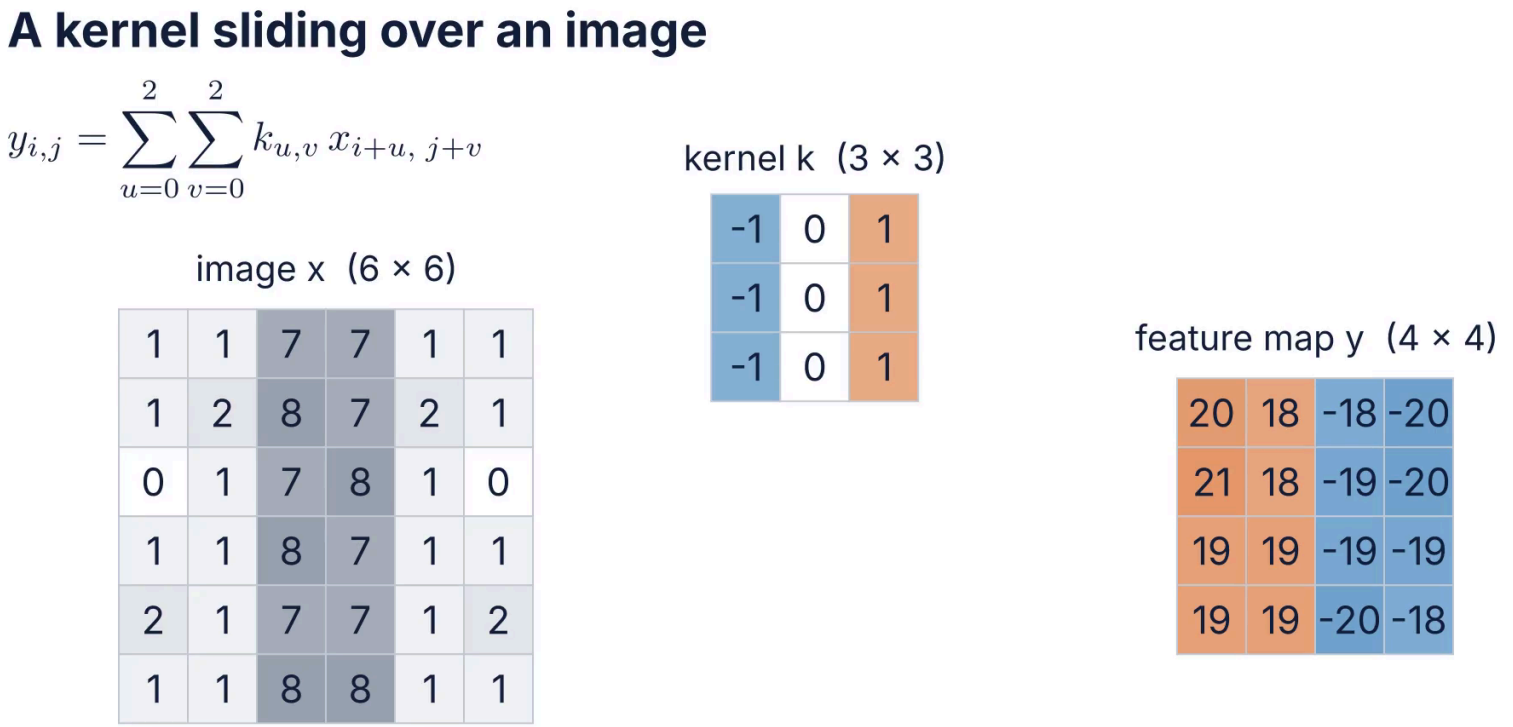
Convolution as a sliding dot product

A kernel $k \in \mathbb{R}^{K \times K}$ slides over the image:

$$y_{i,j} = b + \sum_{u=0}^{K-1} \sum_{v=0}^{K-1} k_{u,v}\, x_{i+u,\; j+v}$$

Deep learning libraries compute this cross-correlation and call it convolution. The kernel is learned, so the flip does not matter.

A kernel sliding over an image



Open animation

A kernel is a pattern detector

The output is large where the patch **looks like** the kernel.

$$\begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix} \quad \text{responds to vertical edges}$$

The result y is a **feature map**: where in the image the pattern occurs.

input x

$1 \times 128 \times 128$



vertical edges

-1	0	1
-2	0	2
-1	0	1

$\downarrow \quad y = k \star x$



horizontal edges

-1	-2	-1
0	0	0
1	2	1

$\downarrow \quad y = k \star x$



blur

$\frac{1}{9}$	$\frac{1}{9}$	$\frac{1}{9}$
$\frac{1}{9}$	$\frac{1}{9}$	$\frac{1}{9}$
$\frac{1}{9}$	$\frac{1}{9}$	$\frac{1}{9}$

$\downarrow \quad y = k \star x$



sharpen

0	-1	0
-1	5	-1
0	-1	0

$\downarrow \quad y = k \star x$



orange: positive response · blue: negative

Same image, same sliding operation: the kernel decides which pattern lights up. In a CNN the kernels are learned.

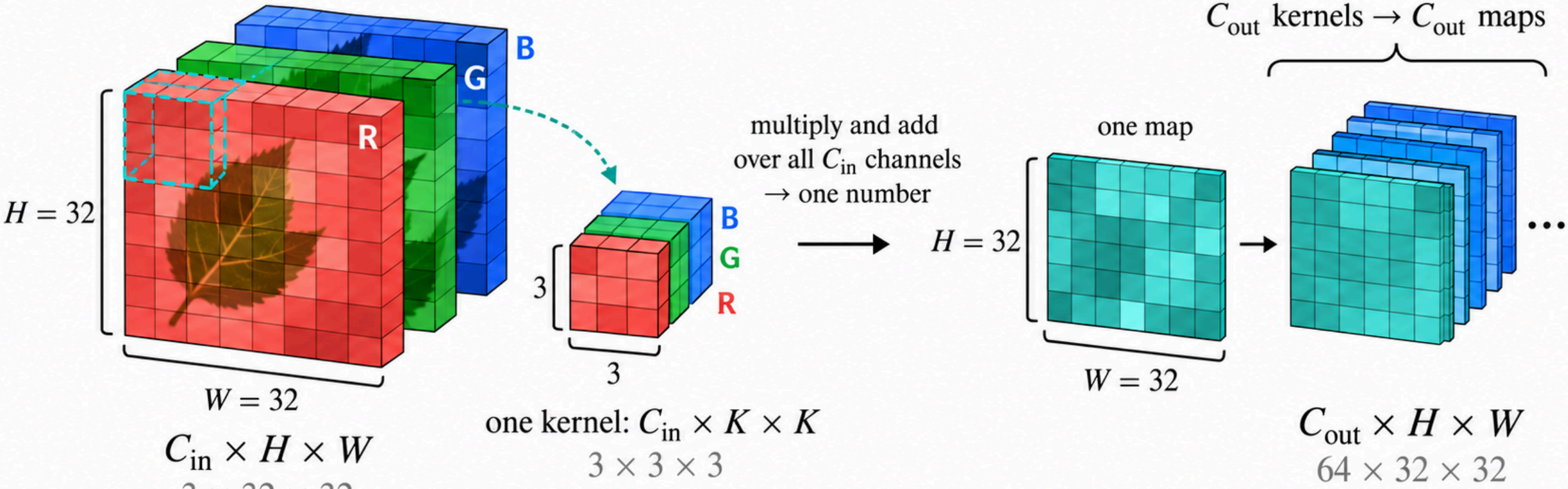
Many channels in, many channels out

An RGB input has 3 channels; each kernel spans **all** input channels.

A layer with C_{out} kernels produces C_{out} feature maps:

$$W \in \mathbb{R}^{C_{\text{out}} \times C_{\text{in}} \times K \times K}, \qquad \#\text{params} = C_{\text{out}}\, C_{\text{in}}\, K^2 + C_{\text{out}}$$

`nn.Conv2d(3, 64, kernel_size=3)` $\rightarrow 64 \cdot 3 \cdot 9 + 64 = 1792$ parameters.



$$\#\text{params} = C_{\text{out}} \cdot C_{\text{in}} \cdot K^2 + C_{\text{out}} = 64 \cdot 3 \cdot 9 + 64 = 1792$$

Padding and stride

Padding p adds a border so the kernel can sit on edge pixels.

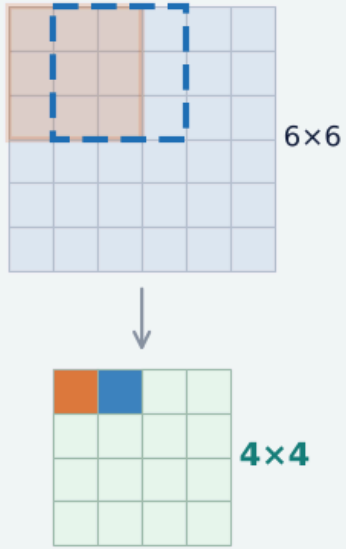
Stride s moves the kernel s pixels at a time.

$$H_{\text{out}} = \left\lfloor \frac{H + 2p - K}{s} \right\rfloor + 1$$

$K = 3,\ p = 1,\ s = 1$ keeps the size: the usual choice inside a block.

no padding

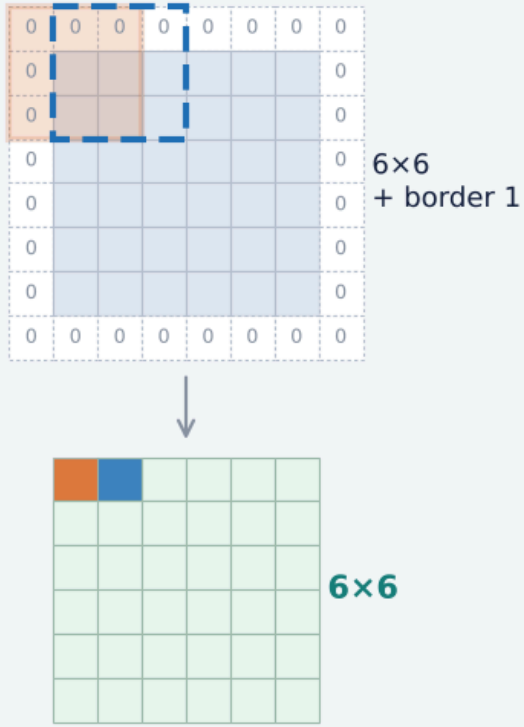
$K = 3, \quad p = 0, \quad s = 1$



$H_{out} = \lfloor (6 + 2 \cdot 0 - 3) / 1 \rfloor + 1 = 4$

padding 1: size kept

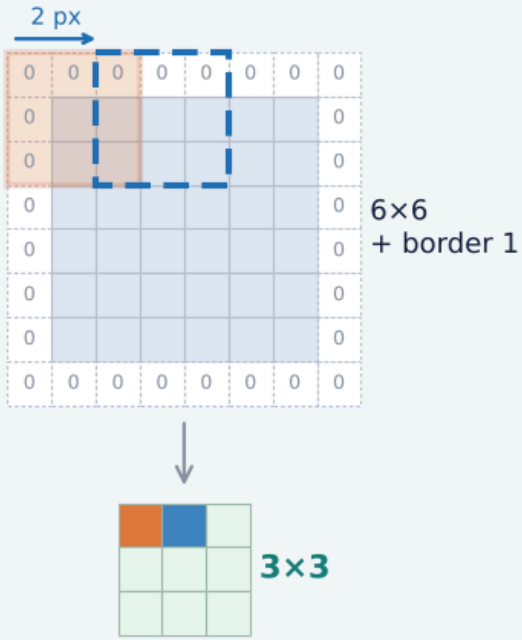
$K = 3, \quad p = 1, \quad s = 1$



$H_{out} = \lfloor (6 + 2 \cdot 1 - 3) / 1 \rfloor + 1 = 6$

padding 1, stride 2: size halved

$K = 3, \quad p = 1, \quad s = 2$



$H_{out} = \lfloor (6 + 2 \cdot 1 - 3) / 2 \rfloor + 1 = 3$

Padding lets the kernel sit on the edge pixels. Stride decides how far it moves: stride 2 halves the map.

Prediction before experiment

Input: $64 \times 32 \times 32$.

Layer: `nn.Conv2d(64, 128, kernel_size=3, stride=2, padding=1)`.

1. What is the output shape?
2. How many parameters?
3. Did the number of activations go up or down?

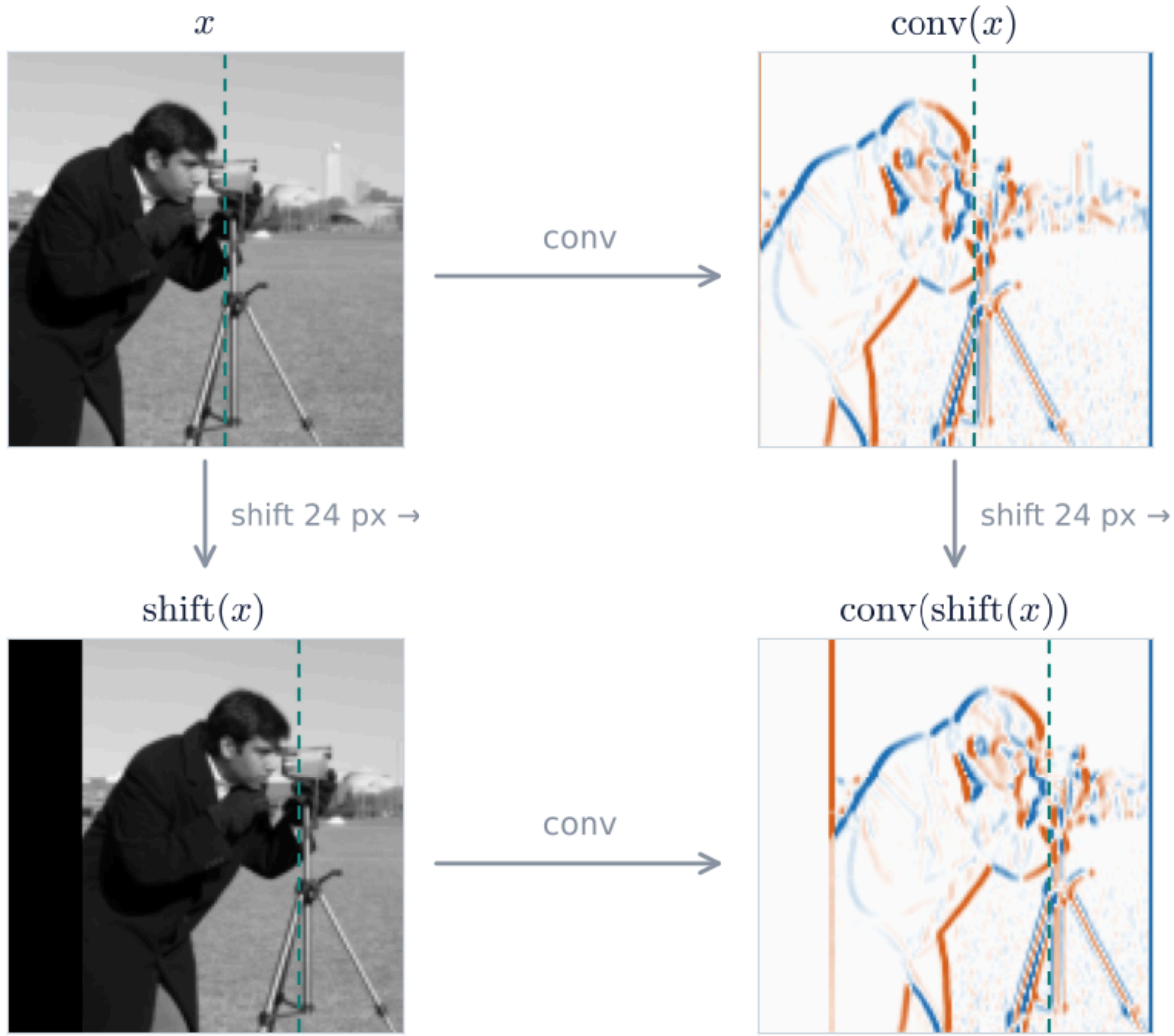
Equivariance

Shift the input, and the feature map shifts by the same amount:

$$\mathrm{conv}\left(\mathrm{shift}(x)\right) = \mathrm{shift}\left(\mathrm{conv}(x)\right)$$

The layer does not need to learn the same pattern at every position.

Equivariance is exact away from the borders, and stride breaks it partially.



Shift then convolve, or convolve then shift: the same feature map.

Difference: exactly 0, except in the columns next to the borders, where padding and the 24 new zero columns enter.

dashed line: the same image column, before and after the shift

$|\mathrm{conv}(\mathrm{shift}(x)) - \mathrm{shift}(\mathrm{conv}(x))|$



A pattern learned in one place is detected everywhere: the layer moves with the image.

Pooling

Pooling summarizes a neighborhood with one number:

- **max pooling**: is the pattern present anywhere in the window?
- **average pooling**: how much of it is there?

It halves the resolution and gives a small **invariance** to local shifts.

Modern networks often replace pooling with a strided convolution.

feature map

1	0	**7**	1
9	1	0	0
0	0	1	0
6	1	**8**	2

max pooling

9	7
6	8

average pooling

2.75	2.00
1.75	2.75

the same map, shifted 1 px right

0	1	0	**7**
0	**9**	1	0
0	0	0	1
0	**6**	1	**8**

max pooling

9	7
6	8

average pooling

2.50	2.00
1.50	2.50

max pooling: identical after the shift · average pooling: 3 of 4 values change

Pooling halves the resolution. A response that moves inside its window does not move the max.

Receptive field

The **receptive field** of a unit is the region of the input it depends on.

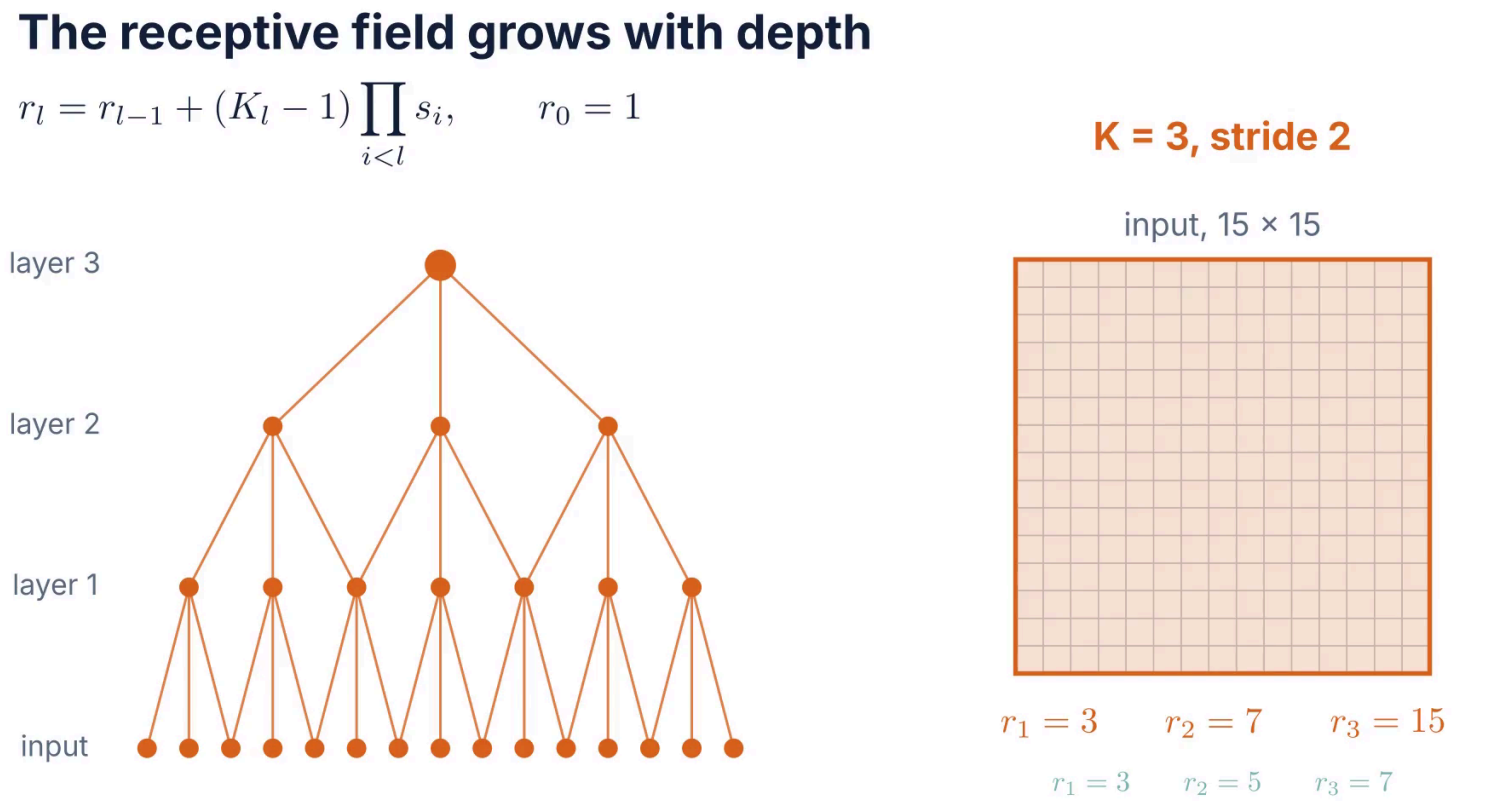
It grows with depth:

$$r_l = r_{l-1} + (K_l - 1) \prod_{i<l} s_i, \qquad r_0 = 1$$

Three 3×3 layers with stride 1 see 7×7. Strides make it grow much faster.

Padding does not change r_l; it only keeps the map size.

The receptive field grows with depth



Stride 2: each layer adds 2 × (product of earlier strides). Three layers see all 15 × 15.

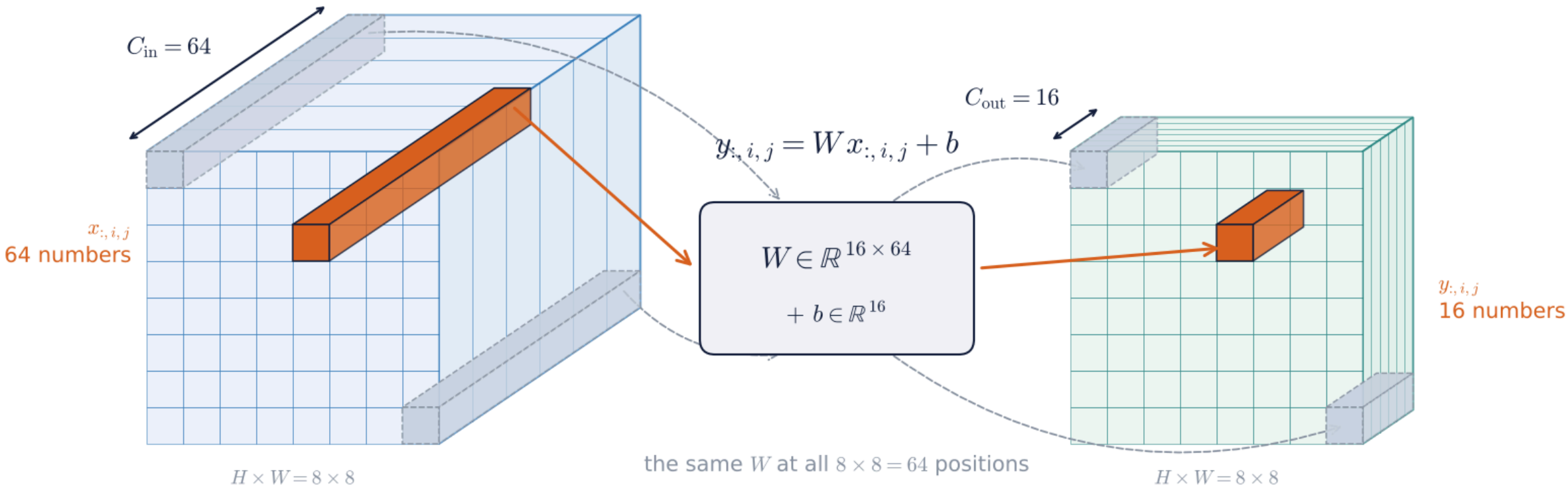
Open animation

1×1 convolutions

A 1×1 kernel looks at a single position, across all channels:

$$y_{:,i,j} = W\, x_{:,i,j} + b, \qquad W \in \mathbb{R}^{C_{\text{out}} \times C_{\text{in}}}$$

It is a small MLP applied to every pixel. It mixes channels and changes their number cheaply.



parameters: $16 \cdot 64 + 16 = 1040$, whatever H and W

A 1 × 1 convolution is a small dense layer applied to every pixel: it mixes channels, not positions.

Parameters versus compute

A convolution is cheap in **parameters** but not in **compute**:

$$\text{FLOPs} \approx 2\, C_{\text{out}}\, C_{\text{in}}\, K^2\, H_{\text{out}}\, W_{\text{out}}$$

The same 1792 weights are reused at every one of the $H_{\text{out}} W_{\text{out}}$ positions.

This is why CNNs reduce resolution as they add channels.

Part 3 · CNNs and residual networks

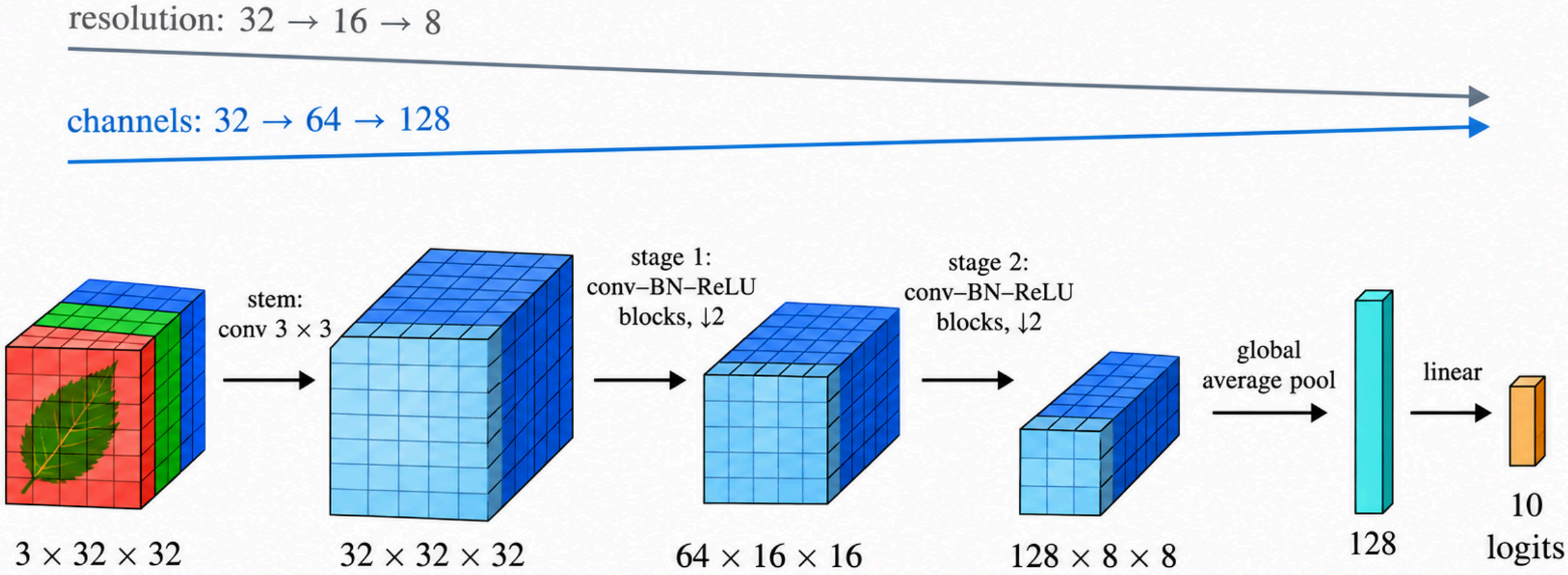
1. Images are structured data — what an MLP ignores
2. Convolution — locality and weight sharing as a layer
3. **CNNs and residual networks** — stacking local layers into a hierarchy
4. Transfer learning — reusing features learned on other data
5. From patches to attention — a weaker assumption, more data

> Part 2: a convolution detects one local pattern everywhere; depth grows what each unit sees.

Anatomy of a CNN

$$\underbrace{\text{stem}}_{\text{first conv}} \rightarrow \underbrace{\text{stage}_1 \rightarrow \cdots \rightarrow \text{stage}_S}_{\text{conv–BN–ReLU blocks}} \rightarrow \underbrace{\text{global pool}}_{C\times 1\times 1} \rightarrow \underbrace{\text{linear}}_{\text{logits}}$$

Between stages: **halve** the resolution, **double** the channels.


resolution: 32 → 16 → 8
channels: 32 → 64 → 128
stem:
conv 3 × 3
stage 1:
conv–BN–ReLU
blocks, ↓2
stage 2:
conv–BN–ReLU
blocks, ↓2
global
average pool
linear
3 × 32 × 32
32 × 32 × 32
64 × 16 × 16
128 × 8 × 8
128
10
logits

A short history in three networks

Network	Year	Idea that stayed
LeNet-5	1998	conv → pool → conv → pool → dense
AlexNet	2012	ReLU, GPUs, Dropout, augmentation, ImageNet scale
VGG	2014	only 3×3 kernels, many of them

After VGG, the problem became **depth** — the same problem as in Session 02.

Why stacks of 3×3?

Two stacked 3×3 layers see 5×5; three see 7×7.

With C channels in and out:

$$\underbrace{2 \cdot 9C^2 = 18C^2}_{\text{two } 3\times3} \quad < \quad \underbrace{25C^2}_{\text{one } 5\times5}$$

Fewer parameters, and one extra nonlinearity in between.

With padding 1 the map stays 32×32: each position sees its own 5×5 window.

VGG-style block

a_l : $64 \times 32 \times 32$

conv 3×3, 64

ReLU

conv 3×3, 64

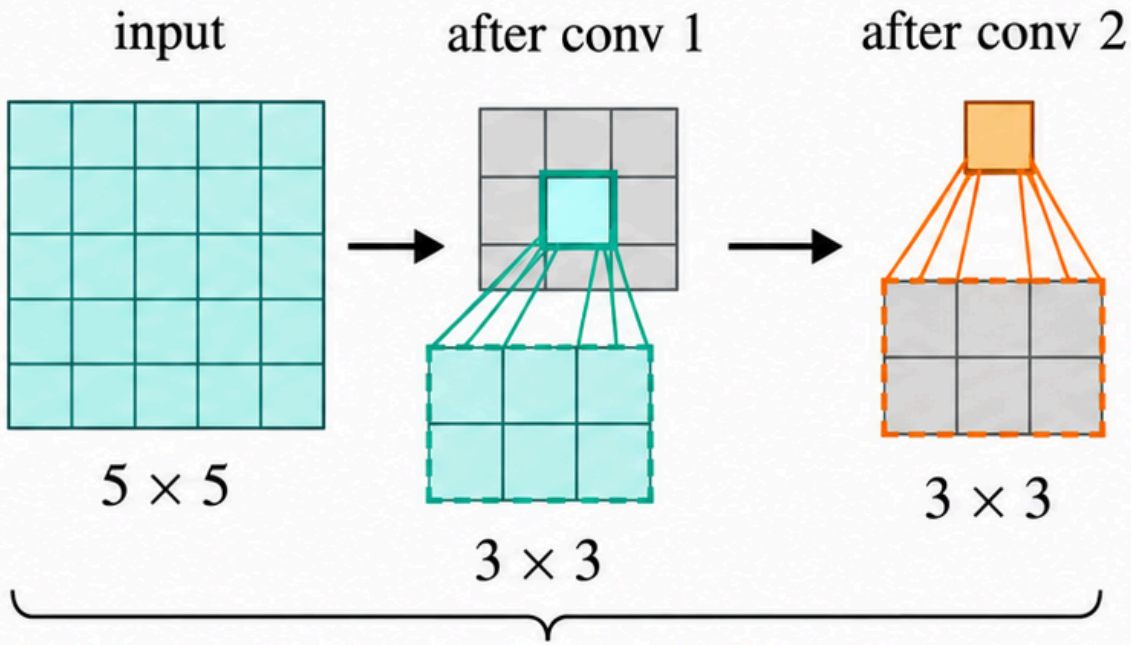
ReLU

max pool 2×2

a_{l+1} : $64 \times 16 \times 16$

plain:
no shortcut

why stack two 3×3?



two 3×3:

$2 \cdot 9C^2 = 18C^2$

+ one extra ReLU

$<$

one 5×5:

$25C^2$

BatchNorm in convolutional layers

In Session 02, BatchNorm normalized each feature over the batch.

In a CNN, each **channel** is one feature, so statistics are computed over N, H and W:

$$\mu_c = \frac{1}{NHW} \sum_{n,i,j} x_{n,c,i,j}$$

`nn.BatchNorm2d(C)` has $2C$ learned parameters, whatever the image size.

Global average pooling

Instead of flattening the last feature maps, average each channel:

$$\bar{a}_c = \frac{1}{HW} \sum_{i,j} a_{c,i,j}$$

- far fewer parameters than `flatten` + dense;
- the network accepts any input size;
- each channel acts as a detector whose overall presence feeds the classifier.

A hierarchy of features

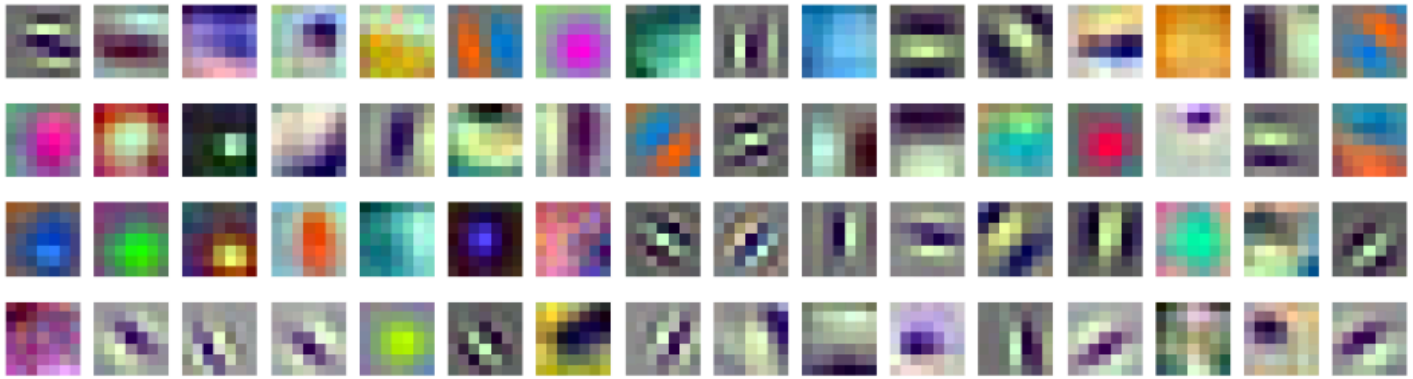
Early layers respond to **edges and colors**.

Middle layers respond to **textures, motifs and small parts** such as eyes.

Late layers respond to **object parts**.

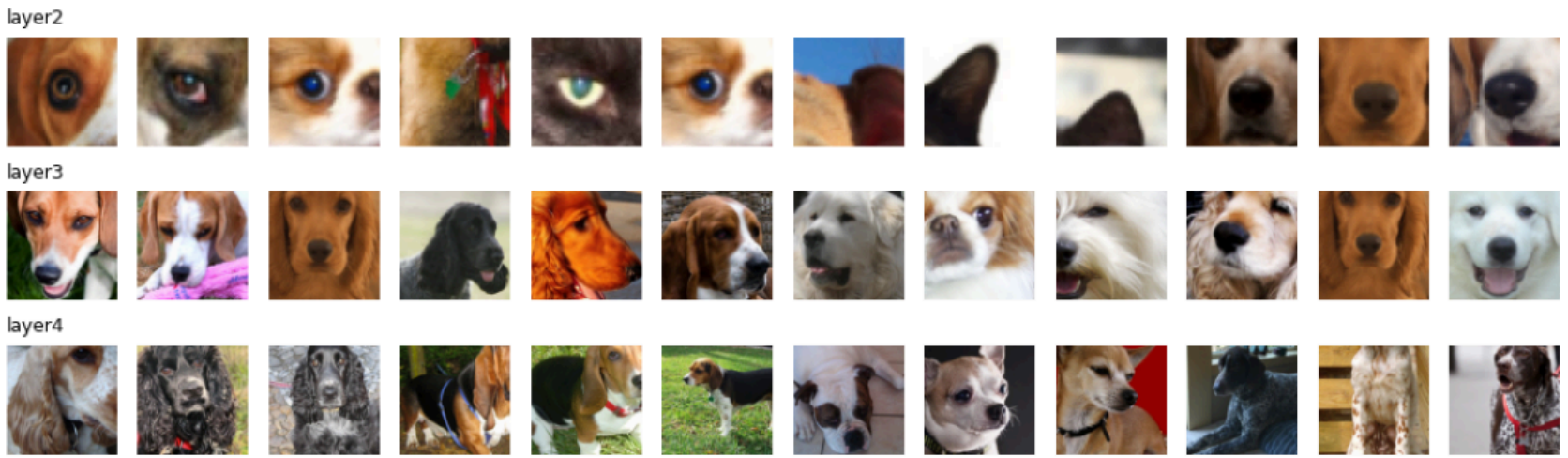
Each level is built from local combinations of the level below.

conv1 · the 64 kernels (7 × 7 × 3) learned on ImageNet



edges at every orientation,
color blobs and color contrasts

patches that most activate 4 channels per stage (3 patches each), Oxford-IIIT Pets



← small parts: eyes, noses, ears

← faces

← whole animals

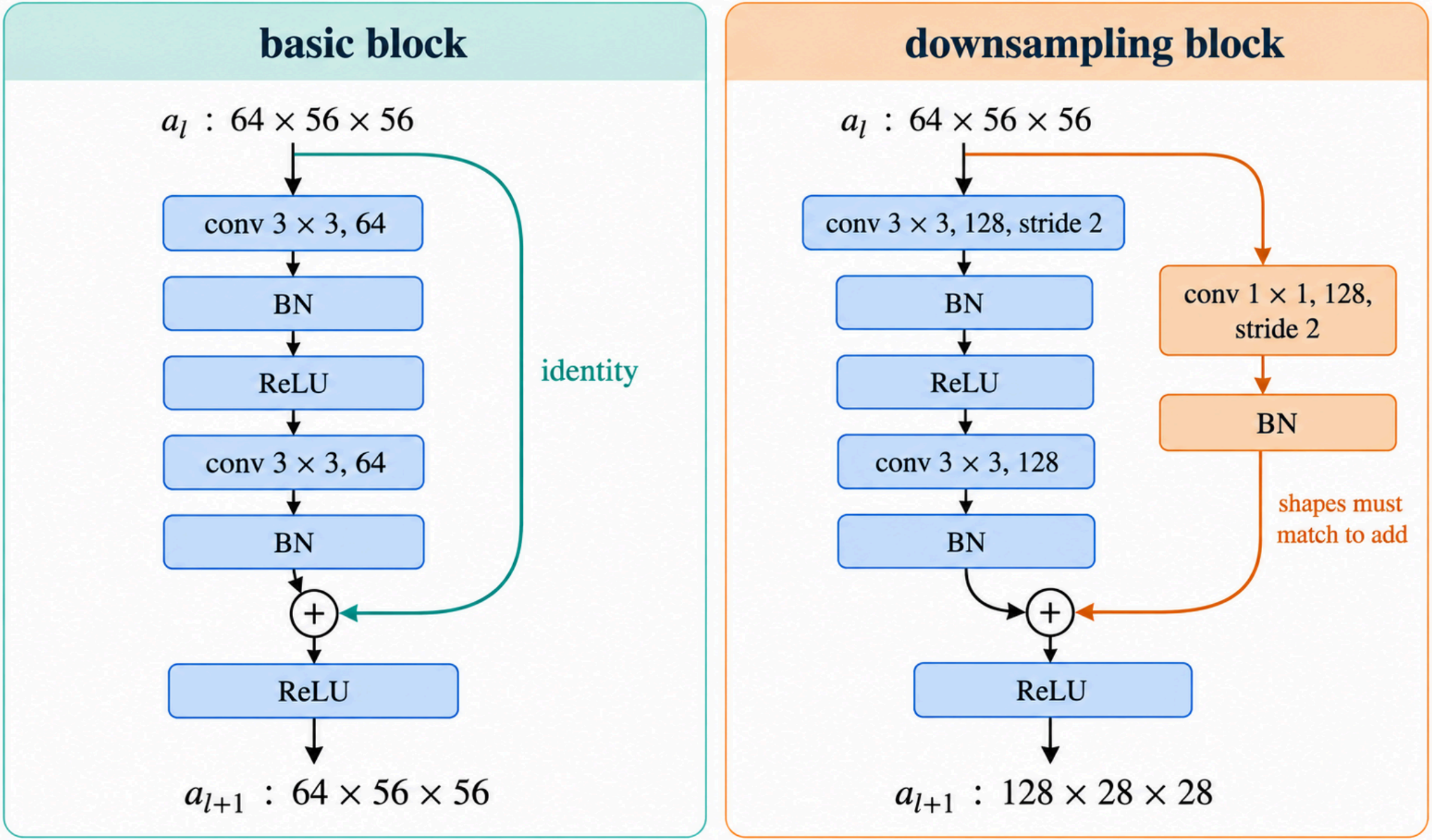
Each stage combines the one below: the receptive field grows, and so does the pattern.

Residual blocks with convolutions

The residual block from Session 02, now with convolutions:

$$a_{l+1} = \mathrm{ReLU}\left(a_l + F(a_l)\right), \qquad F = \mathrm{conv}\text{–}\mathrm{BN}\text{–}\mathrm{ReLU}\text{–}\mathrm{conv}\text{–}\mathrm{BN}$$

When a stage halves the resolution, the shortcut becomes a 1×1 **convolution with stride 2** so the shapes match.


basic block
a_l : 64 × 56 × 56
conv 3 × 3, 64
BN
ReLU
conv 3 × 3, 64
BN
identity
ReLU
a_{l+1} : 64 × 56 × 56
downsampling block
a_l : 64 × 56 × 56
conv 3 × 3, 128, stride 2
BN
ReLU
conv 3 × 3, 128
BN
conv 1 × 1, 128, stride 2
BN
shapes must match to add
ReLU
a_{l+1} : 128 × 28 × 28

ResNet-18 at a glance

Stage	Output (224 input)	Blocks	Channels
stem: 7×7 conv, max pool	56×56	—	64
stage 1	56×56	2	64
stage 2	28×28	2	128
stage 3	14×14	2	256
stage 4	7×7	2	512
global pool + linear	1×1	—	1000 classes

About **11.7 M** parameters — fewer than one dense layer on a 224×224 image.

Augmentation encodes invariances

Random crops, horizontal flips and color jitter tell the model:

> the label does not change under these transformations.

That is another **assumption**. A horizontal flip is harmless for cats and wrong for digits or text.

Augmentation is applied to training data only.

Practice A — From MLP to ResNet

Worked notebooks on CIFAR-10, one change per cell:

- 3.1 · Images and structure: tensors, the permutation experiment, MLP vs small CNN
- 3.2 · Convolution: convolution by hand vs `nn.Conv2d`, kernels, shapes, receptive field
- 3.3 · CNNs and residuals: VGG-style blocks → + BatchNorm → + augmentation → ResNet-20 vs plain-20

> Before each cell: which curve moves, and in which direction?

Part 4 · Transfer learning

1. Images are structured data — what an MLP ignores
2. Convolution — locality and weight sharing as a layer
3. CNNs and residual networks — stacking local layers into a hierarchy
4. **Transfer learning** — reusing features learned on other data
5. From patches to attention — a weaker assumption, more data

> Part 3: a CNN builds a hierarchy of local features; residuals make it deep.

Few labeled images

A real problem rarely has 50 000 labeled images.

With a few dozen per class, a ResNet-18 trained from scratch memorizes the training set.

> Can a network trained on **other** images help with ours?

Features are reusable

Early features (edges, colors, textures) are useful for almost any image.

Late features become specific to the classes of the original task.

So the earlier the layer, the more transferable it is.

pretraining on ImageNet



stem → stage 1 → stage 2 → stage 3 → stage 4 → linear → 1000 classes

general: edges, colors, textures

specific: object parts of the 1000 classes

copy weights

new task: 37 pet breeds

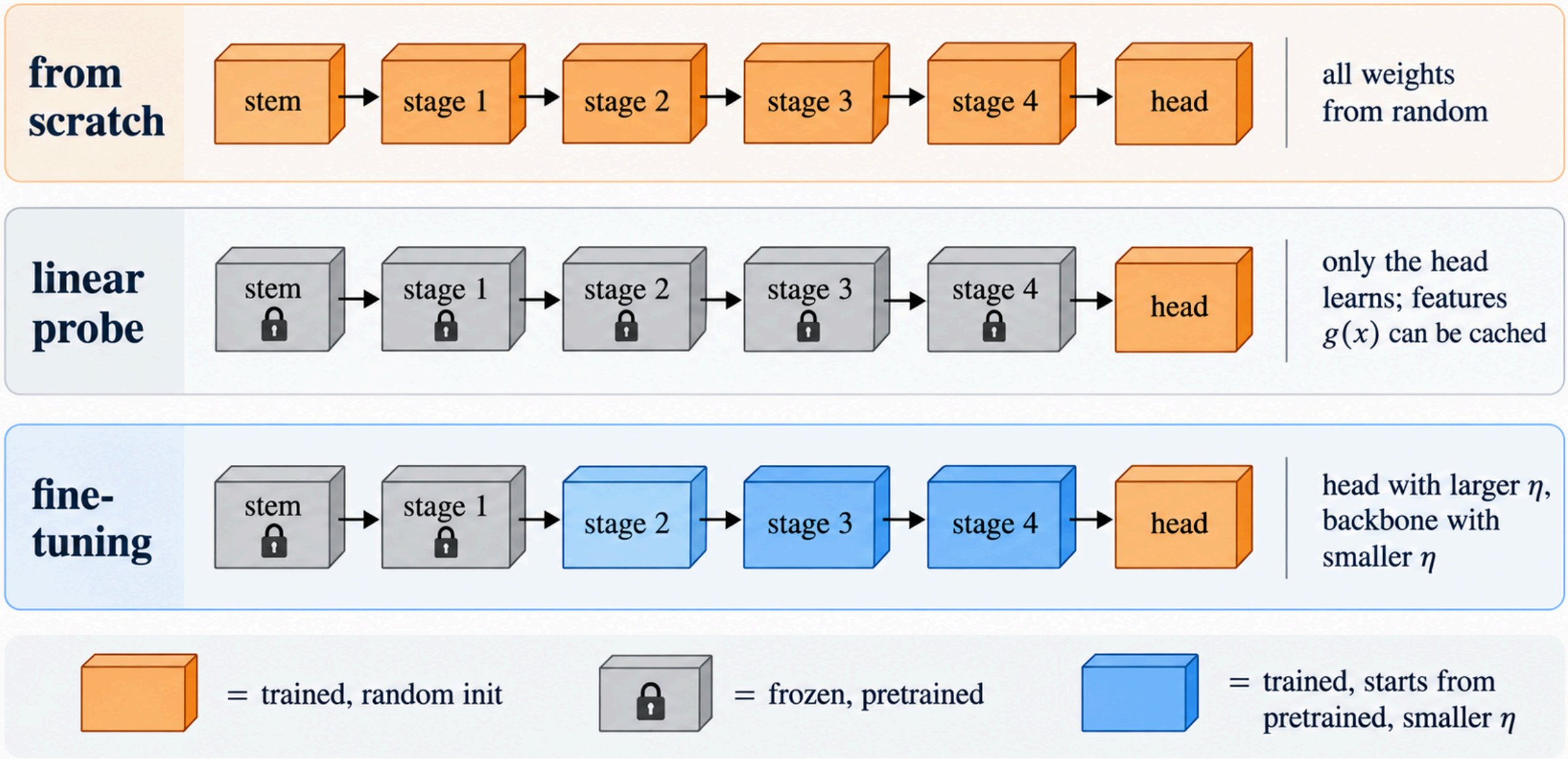


stem → stage 1 → stage 2 → stage 3 → stage 4 → linear → 37 classes

new, random init

Three strategies

Strategy	What is trained	Cost	When
From scratch	everything, random init	high	lots of data, unusual domain
Linear probe	only the new head; backbone frozen	low	little data, similar domain
Fine-tuning	head + some or all backbone	medium	moderate data, or a domain gap


from scratch
stem
stage 1
stage 2
stage 3
stage 4
head
all weights from random
linear probe
stem
stage 1
stage 2
stage 3
stage 4
head
only the head learns; features $g(x)$ can be cached
fine-tuning
stem
stage 1
stage 2
stage 3
stage 4
head
head with larger η, backbone with smaller η
= trained, random init
= frozen, pretrained
= trained, starts from pretrained, smaller η

Linear probe

Freeze the backbone g and train a linear classifier on its features:

$$\hat{y} = \mathrm{softmax}\left(W\, g(x) + b\right), \qquad g \text{ fixed}$$

Features can be computed **once** and cached, so training takes seconds.

Probe accuracy is also a measurement of **representation quality**, which we use again in Session 04.

Fine-tuning

Unfreeze part or all of the backbone, with care:

- a **smaller learning rate** for the backbone than for the new head;
- warmup, so the random head does not wreck pretrained weights in the first steps;
- with small batches, keep BatchNorm statistics **frozen** (`eval` mode) in the backbone.

Match the pretraining preprocessing

The backbone learned on images with a given size and normalization.

```
weights = ResNet18_Weights.IMAGENET1K_V1
model = resnet18(weights=weights)
preprocess = weights.transforms()   # resize, crop, ImageNet mean/std
```

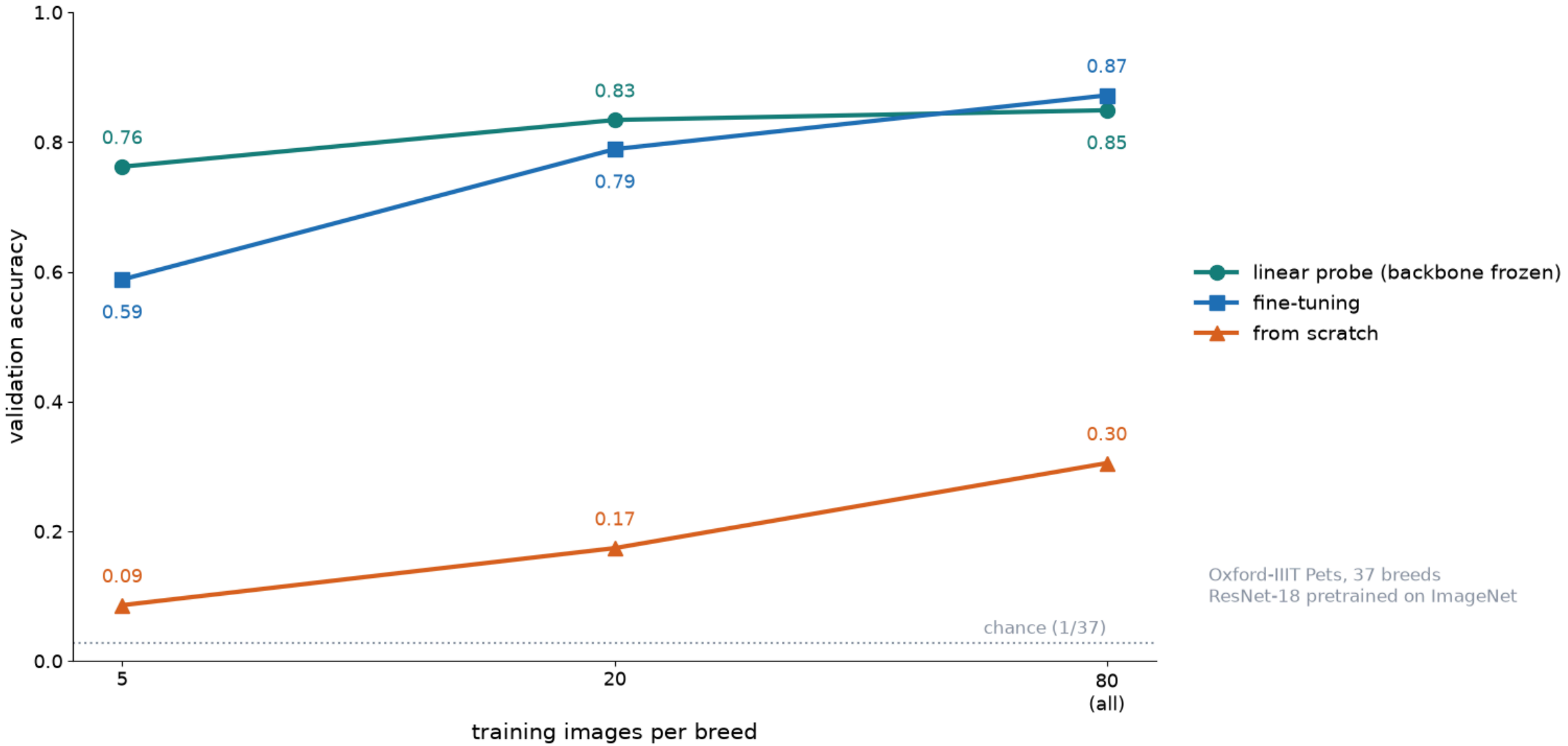
Mismatched preprocessing is one of the most common silent bugs in transfer learning.

Prediction before experiment

Oxford-IIIT Pets, 37 breeds. Three strategies with ResNet-18:

from scratch · linear probe · fine-tuning

1. With **5 images per class**, which wins?
2. With **all** training images (~80 per class), does the order change?



Few labels: the frozen backbone wins. More labels: fine-tuning pulls ahead. From scratch stays far behind.

When transfer helps less

ImageNet is natural photographs taken from the ground.

Satellite images, X-rays and microscopy look different: the **domain gap** is large.

Early features usually still transfer; late features often do not.

Fine-tuning, rather than a linear probe, recovers part of the gap.

Part 5 · From patches to attention

1. Images are structured data — what an MLP ignores
2. Convolution — locality and weight sharing as a layer
3. CNNs and residual networks — stacking local layers into a hierarchy
4. Transfer learning — reusing features learned on other data
5. **From patches to attention** — a weaker assumption, more data

> Part 4: pretrained features let a strong architecture work with few labels.

The limit of locality

A unit in a CNN sees the whole image only after many layers.

Relating two distant regions — a head and a tail, the two ends of a road — takes depth.

> What if every region could look at every other region in **one** layer?

An image as a sequence of patches

Cut a 224×224 image into 16×16 patches:

$$\frac{224}{16} \times \frac{224}{16} = 14 \times 14 = 196 \text{ patches}, \qquad 16 \cdot 16 \cdot 3 = 768 \text{ values each}$$

Each patch is mapped to a vector of size d: a **token**.

Patch embedding is a convolution

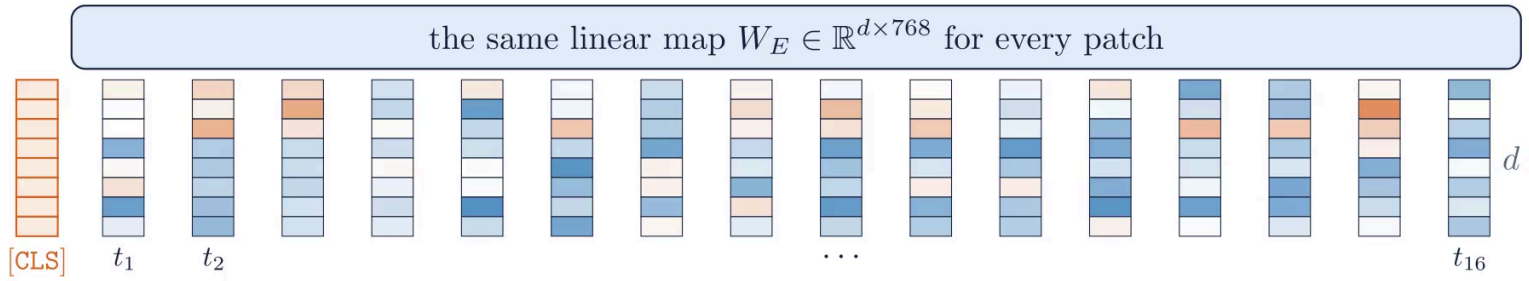
Flattening each patch and applying the same linear map is exactly:

```
nn.Conv2d(3, d, kernel_size=16, stride=16)   # 3×224×224 → d×14×14
```

Then reshape $d \times 14 \times 14$ into a sequence of 196 tokens of size d.

From image to tokens

From image to tokens



17 tokens of size d enter the Transformer

nn.Conv2d(3, d, kernel_size=16, stride=16) computes exactly these tokens.

Open animation

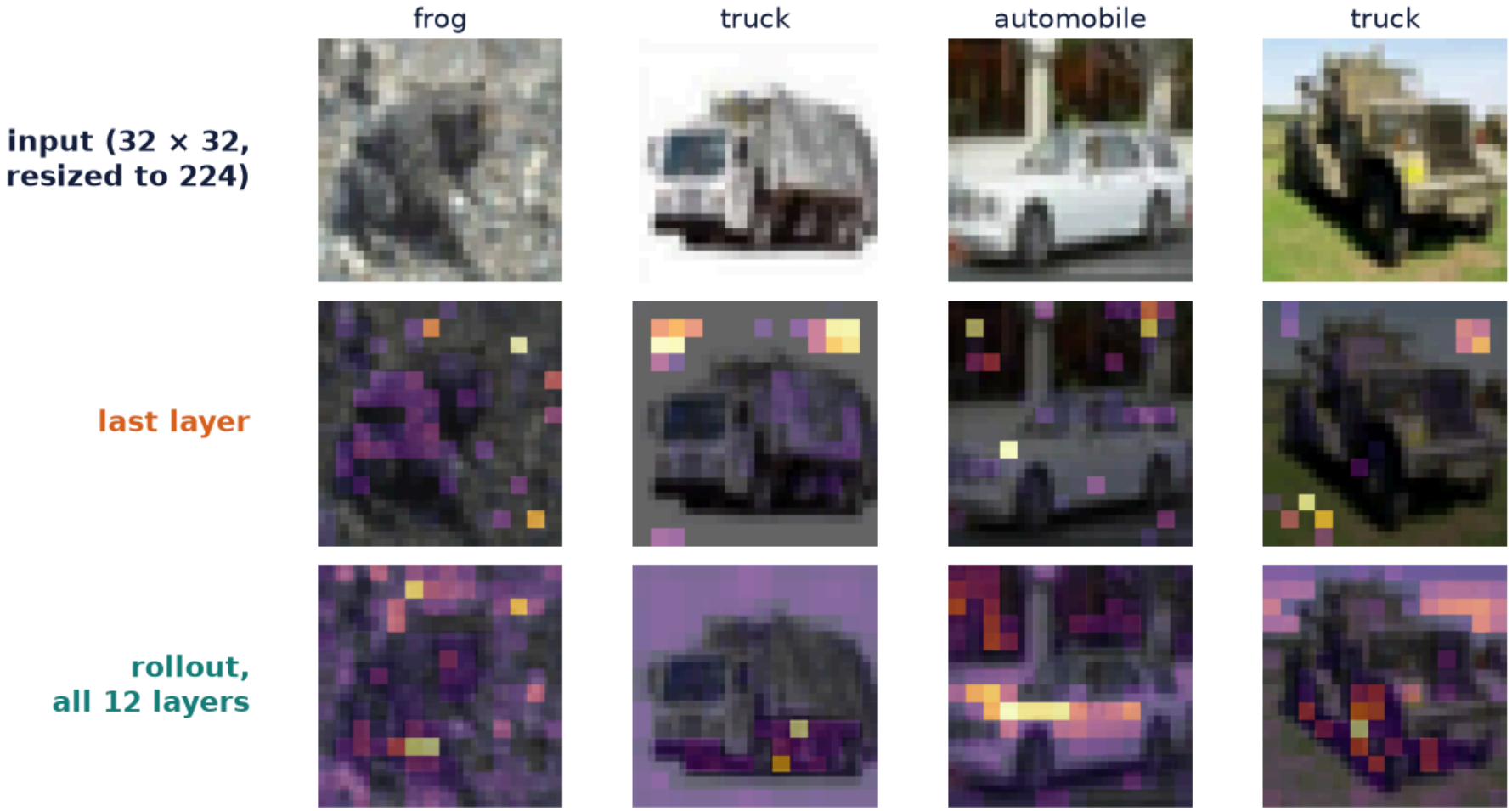
Self-attention, in one slide

Each token becomes a **weighted average of all tokens**. The weights come from how similar the tokens are:

$$\mathrm{Attention}(Q, K, V) = \mathrm{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V, \qquad Q = XW_Q,\ K = XW_K,\ V = XW_V$$

Row i of X is token t_i. The weights depend on the **content**, not on a fixed neighborhood.

The full mechanics — heads, masks, cost — come in **Session 05**.



The last layer parks attention on a few background tokens. Rollout spreads it out. A hint, not an explanation.

Position must be added

Attention compares tokens by content only. Shuffle the tokens, and the outputs are shuffled the same way.

That is the **permutation experiment** of Part 1 again.

So a learned **positional embedding** is added to each token:

$$t_i = W_E \, \text{patch}_i + p_i$$

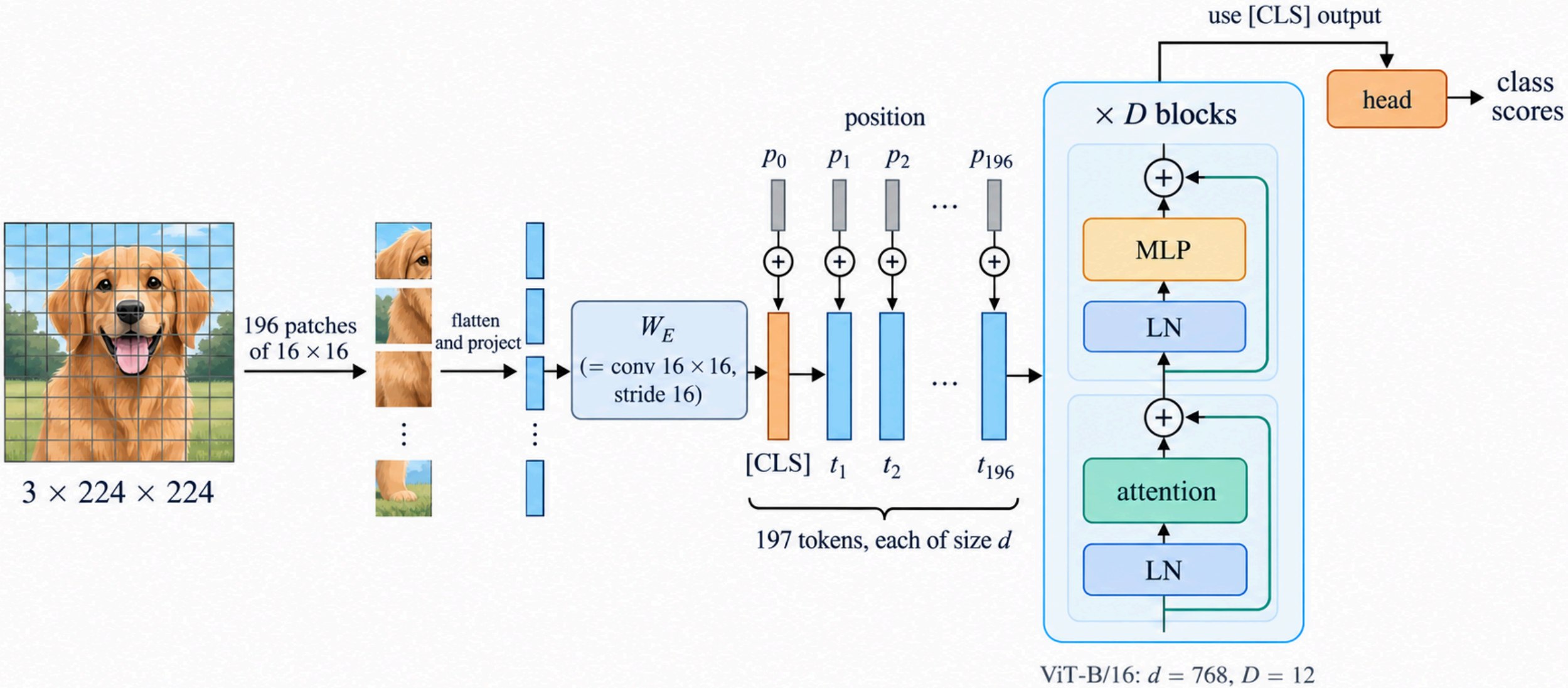
The Vision Transformer

$$[\texttt{CLS}],\ t_1, \ldots, t_{196} \rightarrow \underbrace{\left[\mathrm{LN} \rightarrow \mathrm{attention} \rightarrow + \ \rightarrow \ \mathrm{LN} \rightarrow \mathrm{MLP} \rightarrow +\right]}_{\times D \text{ blocks}} \rightarrow \mathrm{head}(\texttt{CLS})$$

$[\texttt{CLS}]$ is a learned token, not a patch; its output feeds the head.

Everything in the block is from Session 02: LayerNorm, residual paths, an MLP.

The new piece is attention.

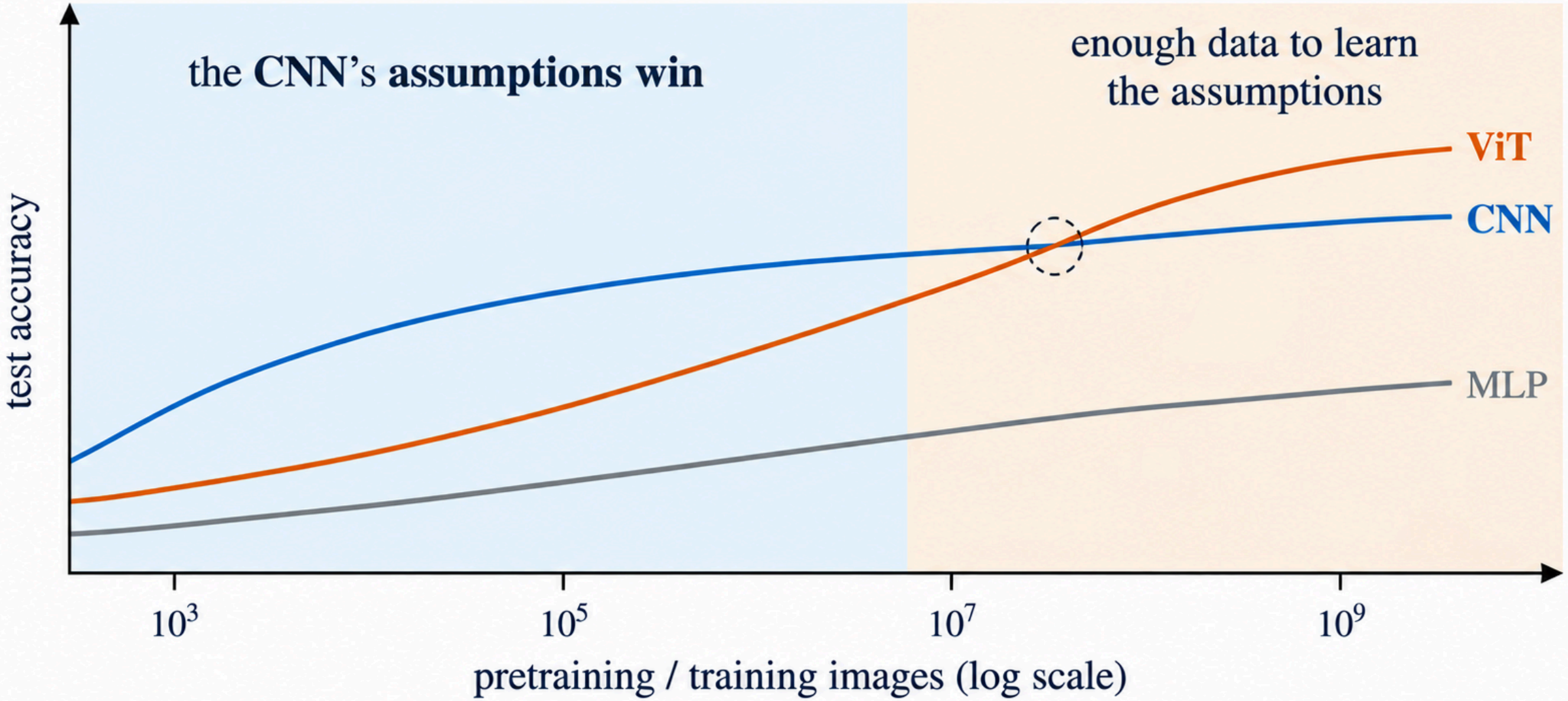

use [CLS] output
head
class scores
× D blocks
position
p_0 p_1 p_2 p_{196}
MLP
LN
attention
LN
196 patches of 16 × 16
flatten and project
W_E (= conv 16 × 16, stride 16)
[CLS] t_1 t_2 t_{196}
197 tokens, each of size d
3 × 224 × 224
ViT-B/16: $d = 768$, $D = 12$

Inductive bias versus data

A CNN **assumes** locality and translation equivariance.

A ViT has to **learn** them from data.

- small data: the CNN's assumptions win;
- very large pretraining: the ViT catches up and then passes the CNN;
- modern CNNs (ConvNeXt) close much of that gap again: the bias is not the whole story.



CNN: assumes locality and translation equivariance

ViT: learns them from data

MLP: assumes nothing about position

schematic, not measured

Prediction before experiment

CIFAR-10, 10 000 training images, same budget of epochs:

1. a small CNN trained from scratch;
2. a small ViT trained from scratch;
3. a pretrained ViT with a linear probe.

Rank them before running.

Practice B — Transfer and attention

- 3.4 · Transfer learning · Oxford-IIIT Pets: from scratch, linear probe and fine-tuning, with 5 to ~80 images per class
- 3.5 · From patches to ViT · CIFAR-10: patch embedding, small ViT vs CNN, no position, pretrained ViT and its attention

> 3.4 and 3.5 run best with a GPU (Colab or Kaggle).

Integrating exercise — EuroSAT

3.6 · Integrating exercise · Sentinel-2 satellite images, 64×64, 10 land-use classes, 27 000 images

- choose and justify a strategy: CNN from scratch, linear probe or fine-tuning;
- preprocessing that matches the backbone, augmentation that respects the domain;
- every run logged to **MLflow**; `predict_raw` on new image files.

> Does ImageNet help with images taken from space?

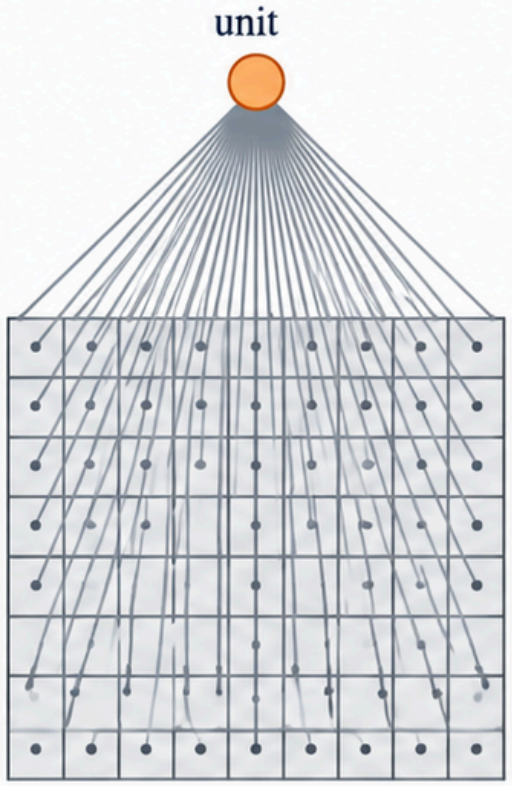
Choosing an architecture

Data	Labels	First choice
tabular features	any	MLP or gradient boosting
images, few labels	~10–1000 per class	pretrained CNN or ViT, linear probe → fine-tune
images, many labels	10^3+ per class	CNN, or ViT with pretraining
images, unusual domain	any	fine-tuning; check the gap with a probe

What each architecture assumes

	Assumes	Pays with
MLP	nothing about position	parameters, data
CNN	locality, translation equivariance	global context needs depth
ViT	almost nothing; position is learned	data or pretraining

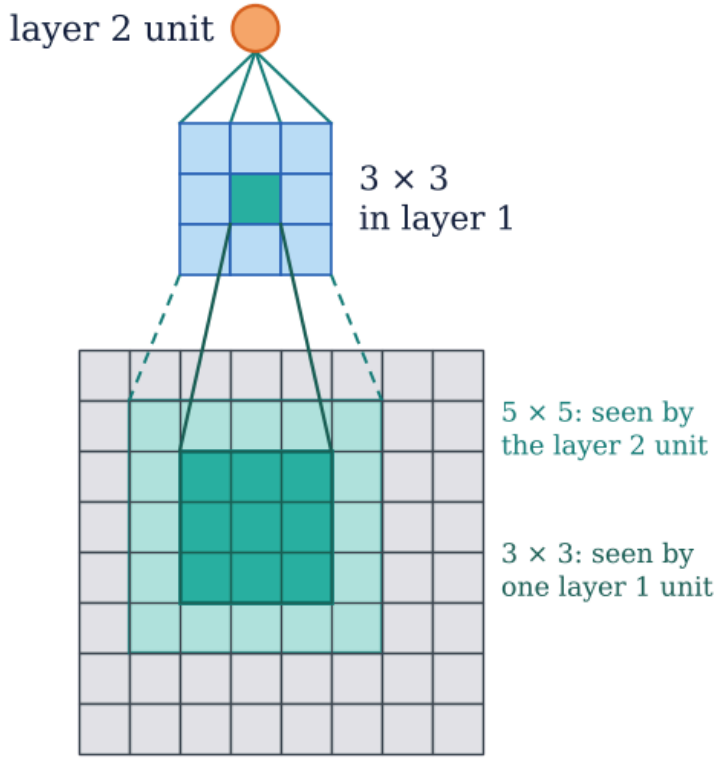
MLP



8 × 8 input (64 cells)

every input, its own weight,
no notion of neighbor

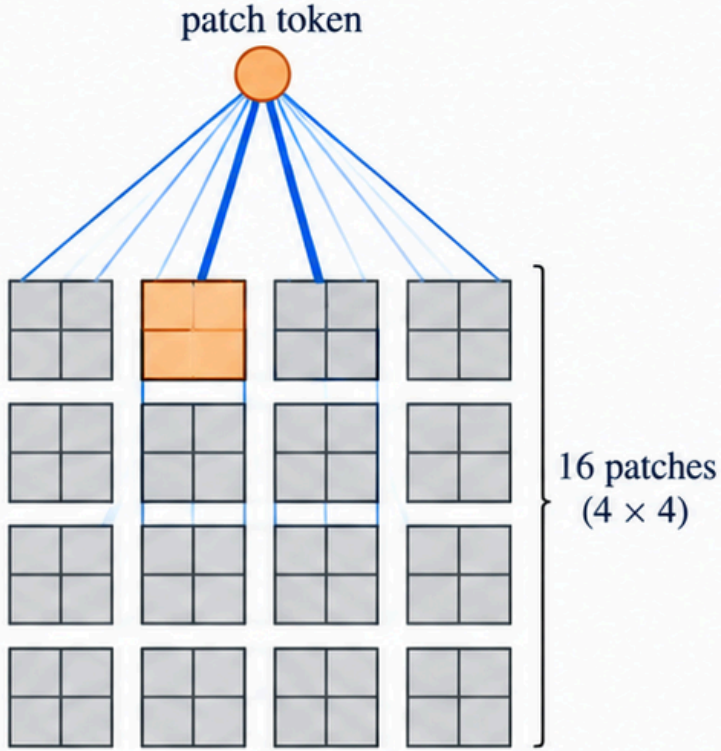
CNN



8 × 8 input (64 cells)

a neighborhood,
growing with depth

ViT



8 × 8 input (split into 2 × 2 patches)

every patch,
weighted by content

What each architecture assumes decides what it must learn from data.

Exit question

You have 300 labeled chest X-rays in 3 classes.

A colleague proposes a ViT-B/16 trained from scratch.

Choose two and justify:

- linear probe on a pretrained CNN;
- fine-tuning a pretrained CNN;
- training a ViT from scratch with heavy augmentation;
- collecting more labels first;
- checking the domain gap with a probe before fine-tuning.